

Chapter 3

Diode Models and Circuits

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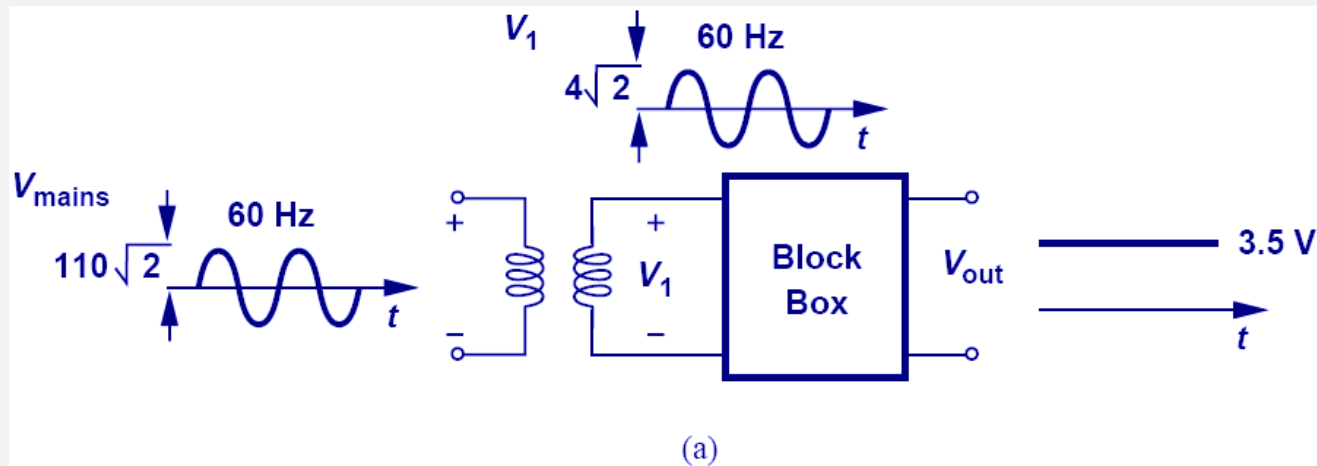
Diode Models and Circuits

- After understanding the basic physics of a diode, we will study its behavior as a circuit element and its many applications in this chapter.

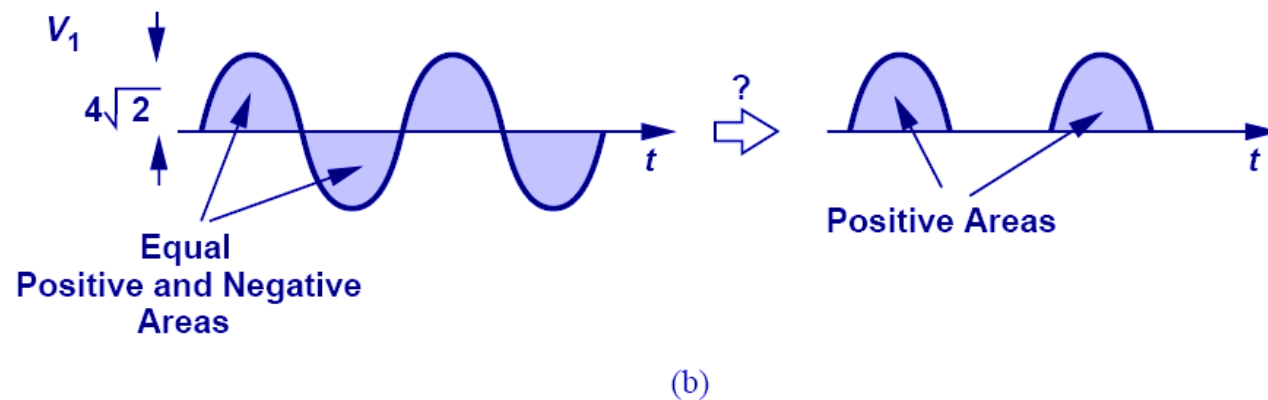


Diode's Application: Cell Phone Charger

- Problem: How to convert 110-V AC voltage to 3.5-V DC voltage?



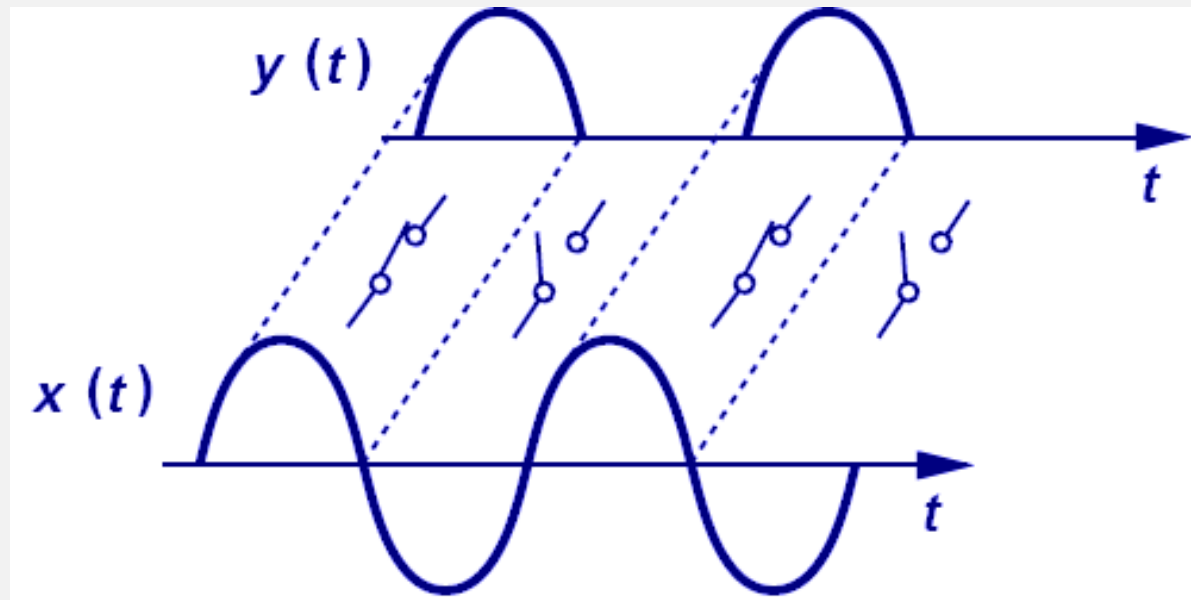
Diode acts as the black box that passes only the positive half of the sinusoid.



(a) Charger circuit, (b) elimination of negative half cycles.

Diode's Action in The Black Box

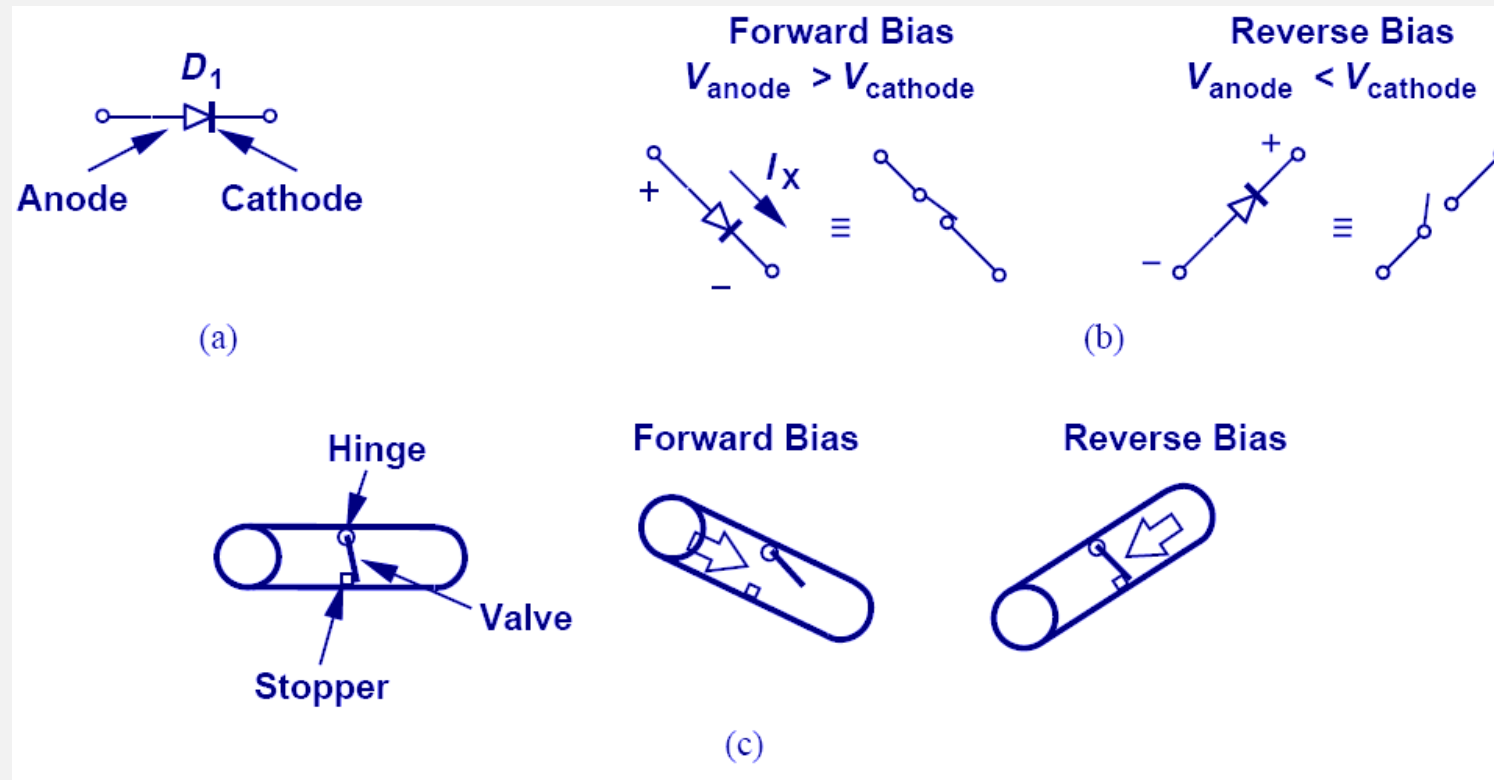
- The diode behaves as a short circuit during the positive half cycle and an open circuit during the negative half cycle.



Conceptual operation of a diode.

Ideal Diode

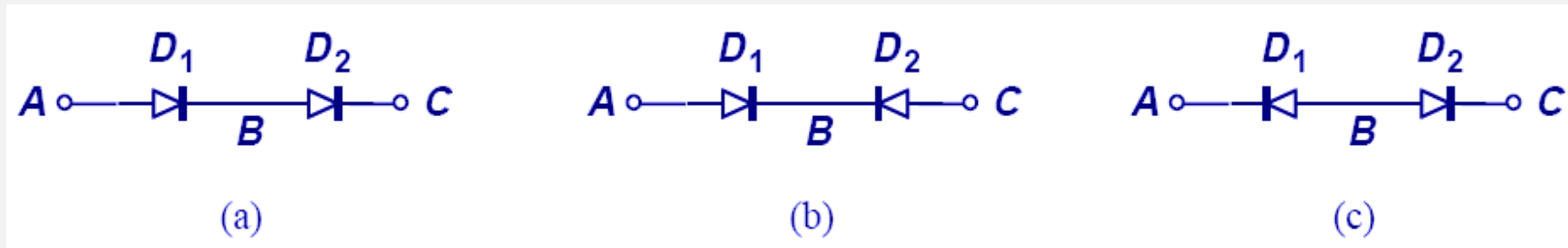
- In an ideal diode, if the voltage across it tends to exceed zero, current flows. It is analogous to a water pipe that allows water to flow in only one direction.



(a) Diode symbol, (b) equivalent circuit, (c) water pipe analogy.

Example 3-1

- As with other two-terminal devices, diodes can be placed in series (or in parallel). Determine which one of the configurations in the figure can conduct current.

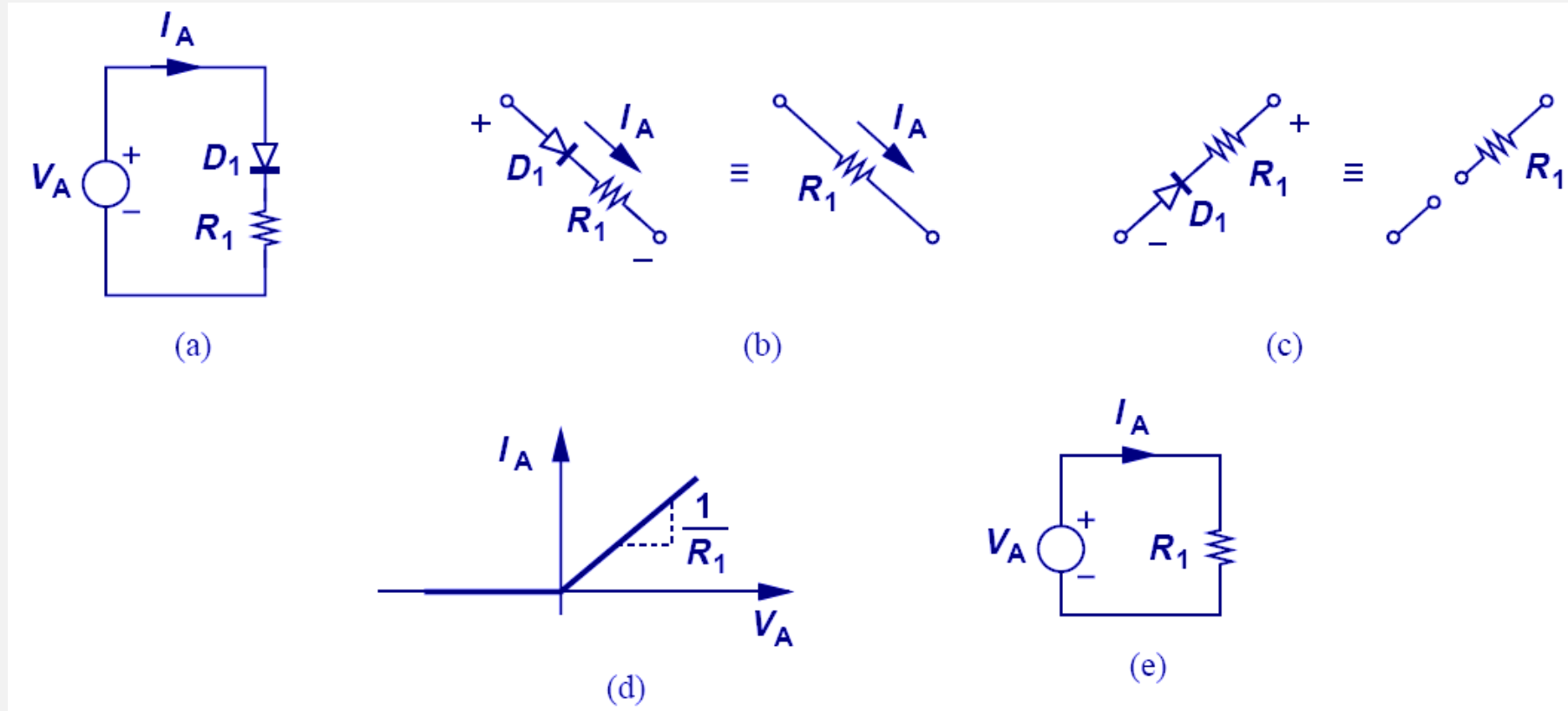


Solution:

Diodes cannot be connected in series randomly. For the circuits above, only (a) can conduct current from A to C.

Diodes in Series

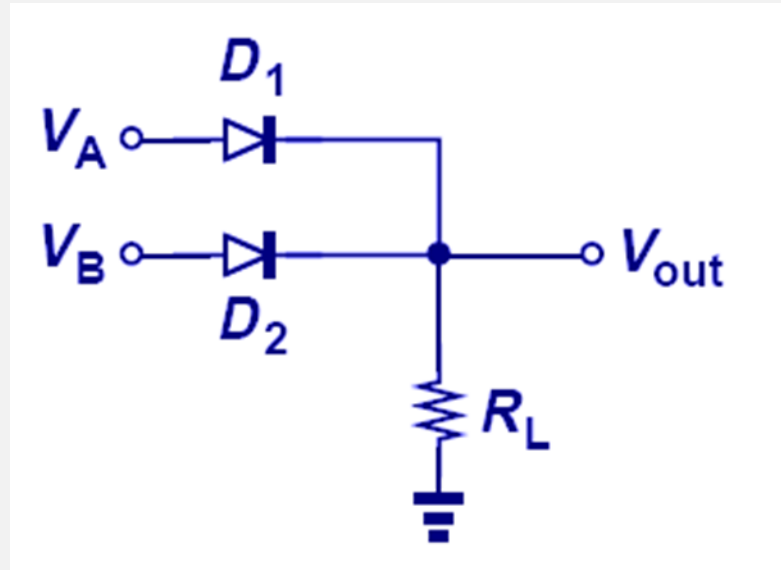
- Plot the I/V characteristic for the diode-resistor combination.



(a) Diode-resistor series combination, (b) equivalent circuit under forward bias, (c) equivalent circuit under reverse bias, (d) I/V characteristic, (e) equivalent circuit if D_1 is on.

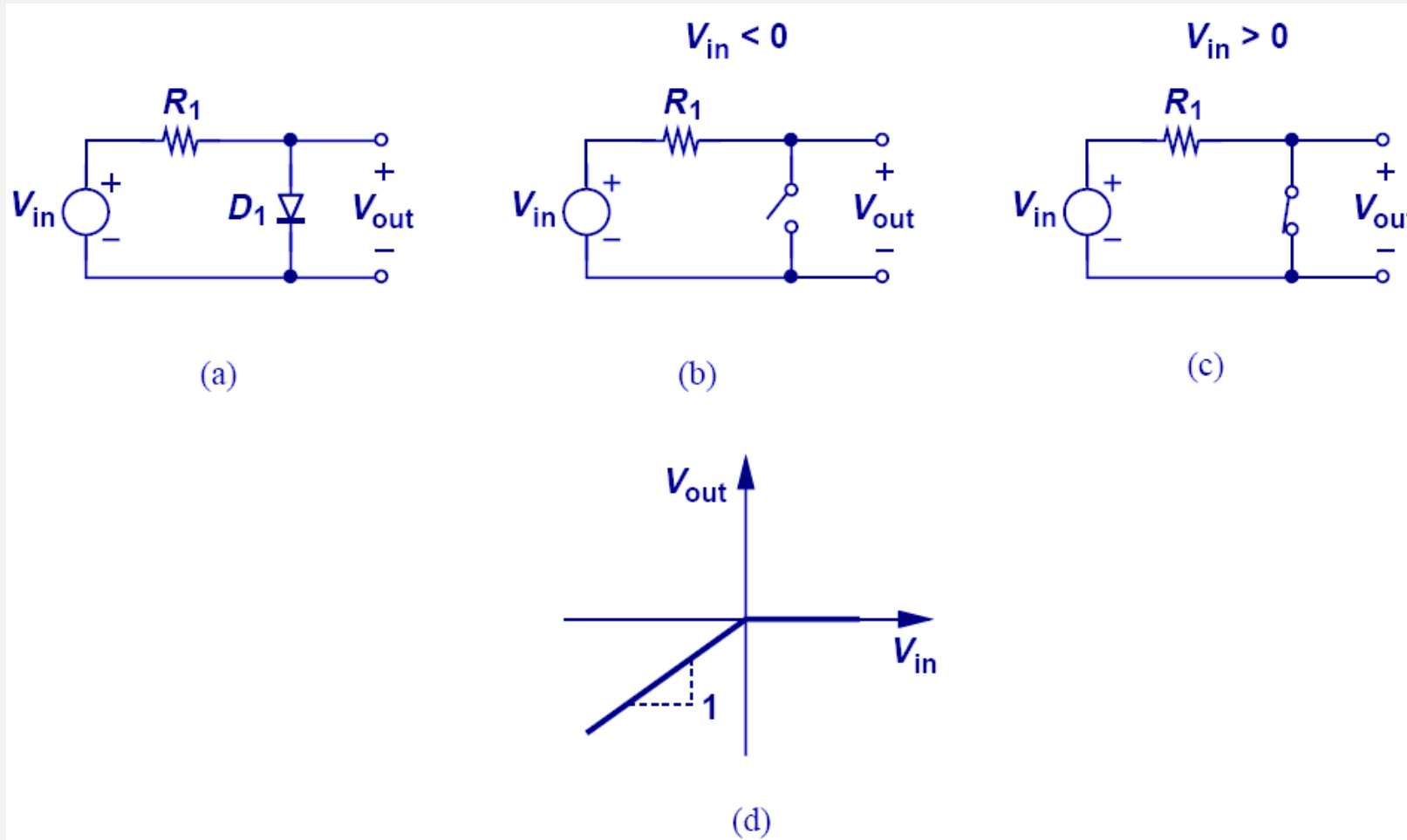
Diode Implementation of OR Gate

- The circuit above shows an example of diode-implemented OR gate.
- V_{out} can only be either V_A or V_B , not both.



- The circuit operates as a logical OR gate and was in fact used in early digital computers.

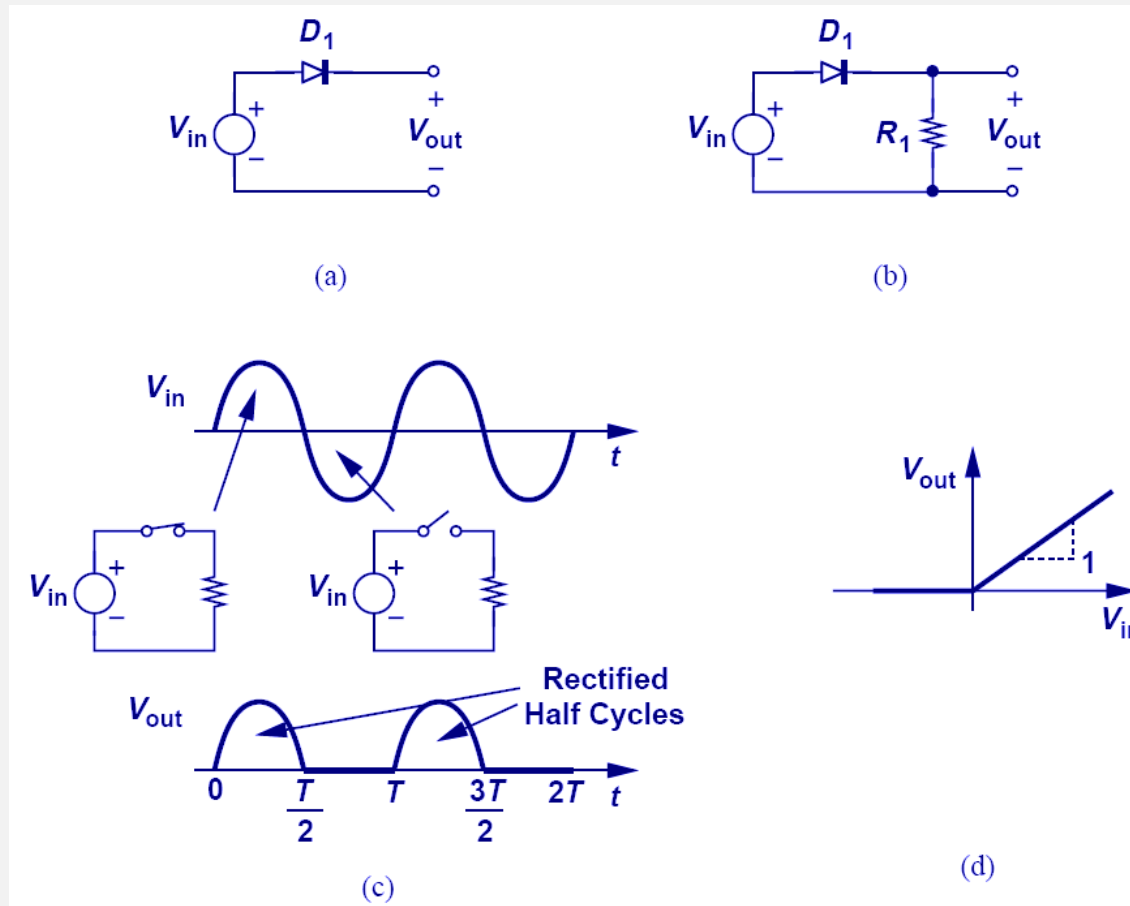
Input/Output Characteristics



(a) Resistor-diode circuit, (b) equivalent circuit for negative input, (c) equivalent circuit for positive input, (d) input/output characteristic.

Diode's Application: Rectifier

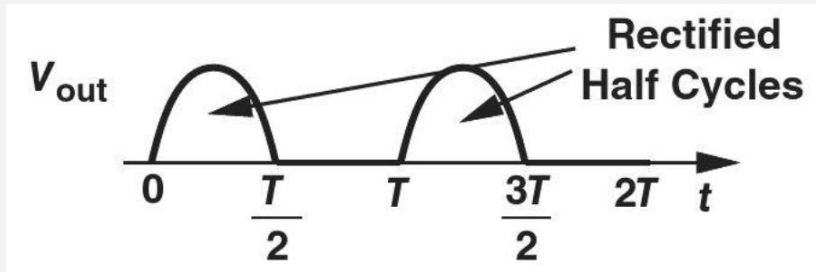
- A rectifier is a device that passes positive-half cycle of a sinusoid and blocks the negative half-cycle or vice versa.



- (a) A diode operating as a rectifier,
- (b) complete rectifier,
- (c) input and output waveforms,
- (d) input/output characteristic.

Diode's Application: Signal Strength Indicator

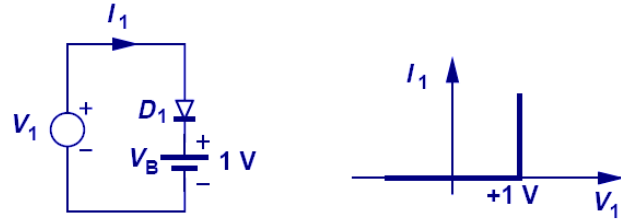
- The averaged value of a rectifier output can be used as a signal strength indicator for the input, since $V_{out,avg}$ is proportional to V_p , the input signal's amplitude.



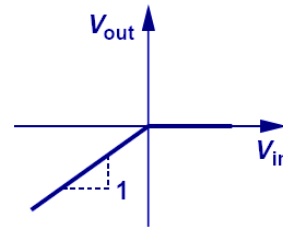
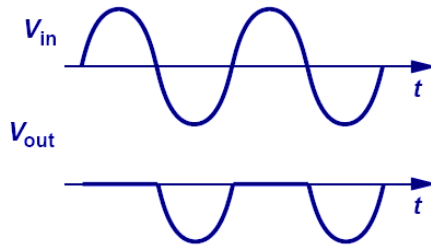
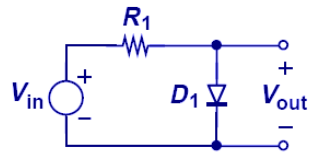
$$V_{out}(t) = \begin{cases} V_p \sin \omega t, & 0 \leq t \leq \frac{T}{2} \\ 0, & \frac{T}{2} \leq t \leq T \end{cases}$$

$$\begin{aligned} V_{out,avg} &= \frac{1}{T} \int_0^T V_{out}(t) dt = \frac{1}{T} \int_0^{T/2} V_p \sin \omega t dt \\ &= \frac{1}{T} \frac{V_p}{\omega} [-\cos \omega t]_0^{T/2} = \frac{V_p}{\pi} \end{aligned}$$

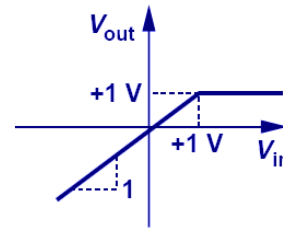
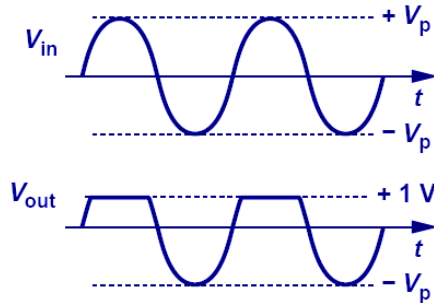
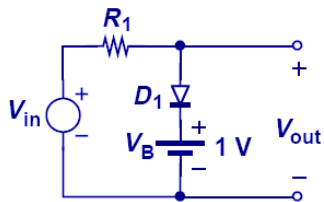
Diode's Application: Limiter



(a)



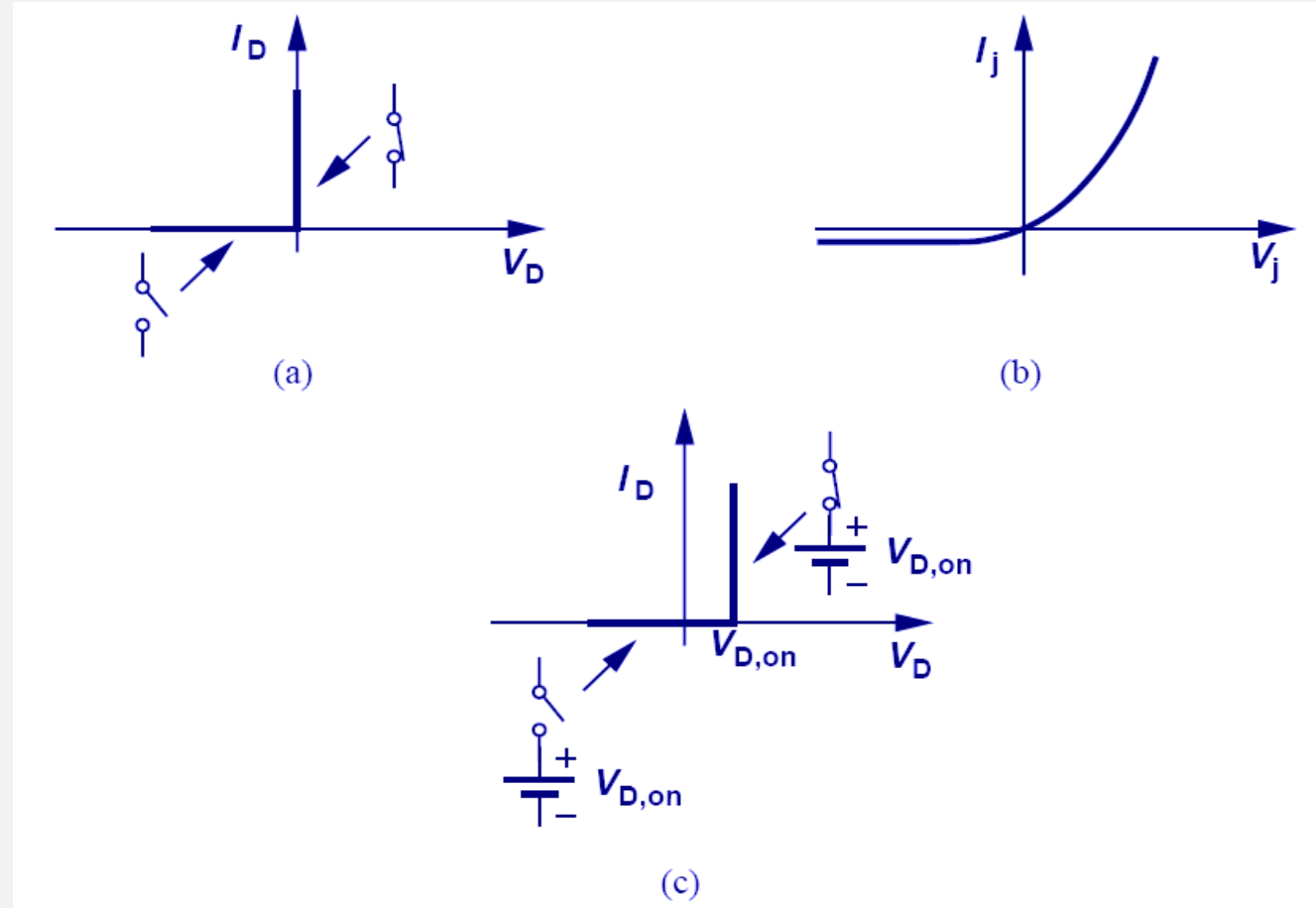
(b)



(c)

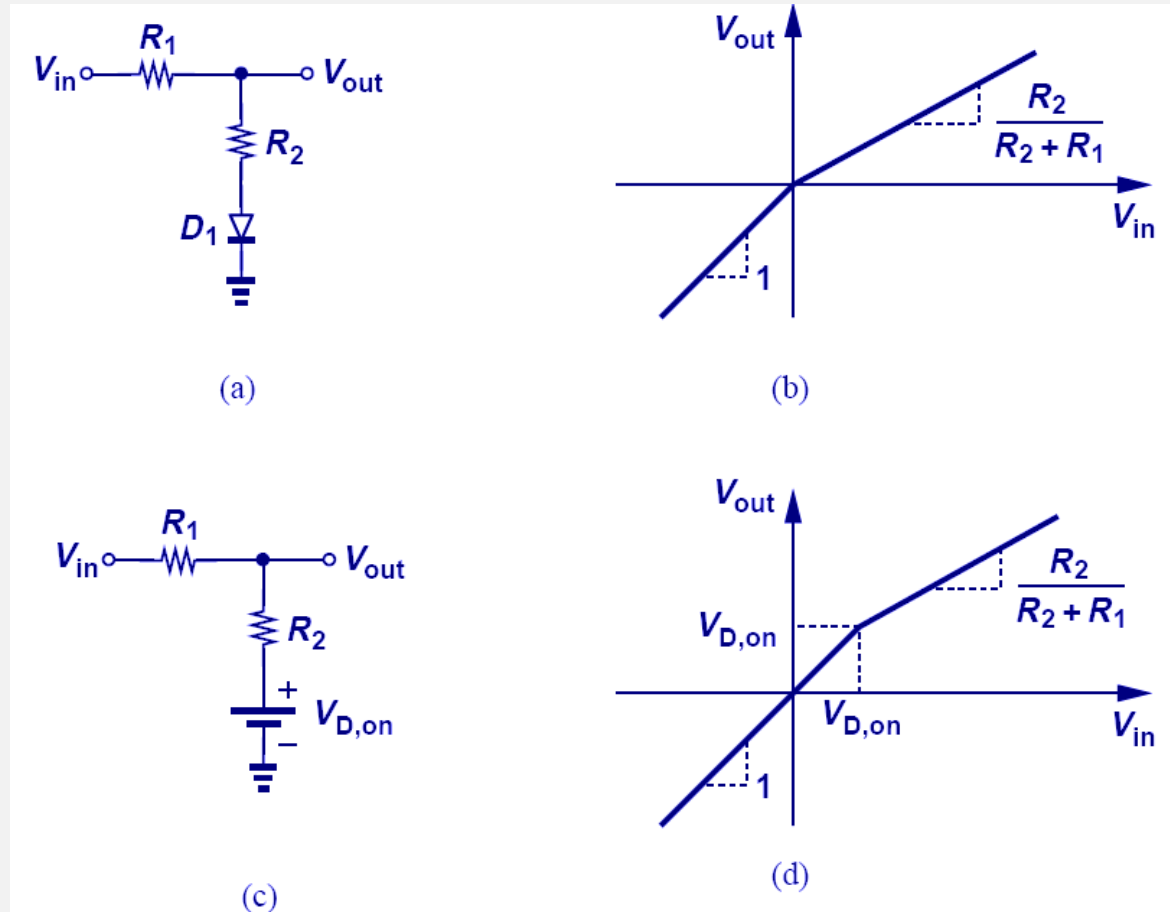
- ✓ The purpose of a limiter is to force the output to remain below certain value.
- ✓ In (a), the addition of a 1 V battery forces the diode to turn on after V_1 has become greater than 1 V.

Different Models for Diode



Diode characteristics: (a) ideal model, (b) exponential model, (c) constant-voltage model.

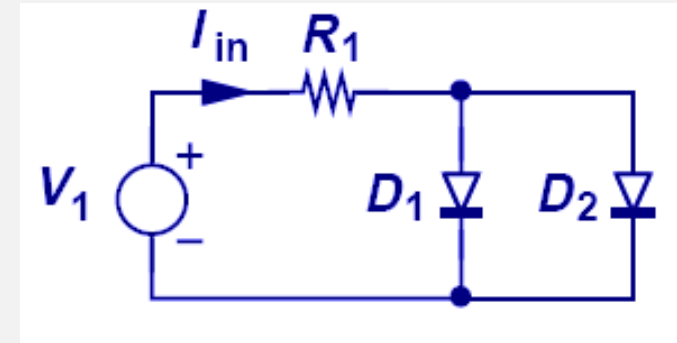
Input/Output Characteristics with Ideal and Constant-Voltage Models



(a) Diode circuit, (b) input/output characteristic with ideal diode model, (c) input/output characteristic with constant-voltage diode model, and (d) circuit with ideal diode model.

Example 3-13: Exponential Model

- Assume D_1 and D_2 have different cross section areas but are otherwise identical. Determine the current flowing through each diode.



Solution:

We must resort to the exponential equation because the ideal and constant-voltage models do not include the device area.

$$I_{in} = I_{D1} + I_{D2}$$

$$I_{D1} = I_{S1} \exp(V_D/V_T), I_{D2} = I_{S2} \exp(V_D/V_T)$$

$$V_D = V_T \ln \frac{I_{D1}}{I_{S1}} = V_T \ln \frac{I_{D2}}{I_{S2}}$$

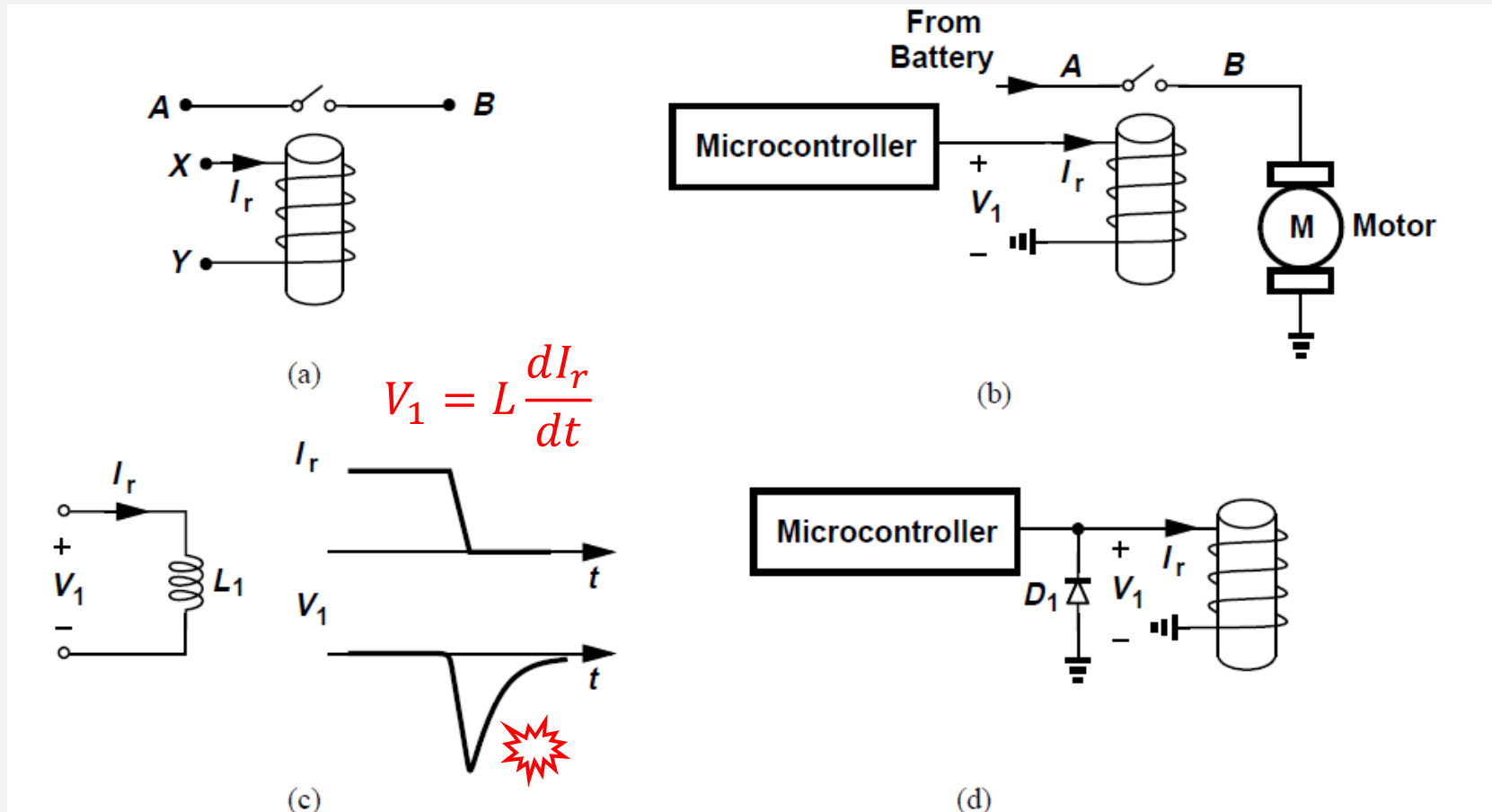
$$\frac{I_{D1}}{I_{S1}} = \frac{I_{D2}}{I_{S2}}$$

$$I_{in} = I_{D1} + I_{D2} = I_{D1} + \frac{I_{D1}}{I_{S1}} I_{S2} = \left(1 + \frac{I_{S2}}{I_{S1}}\right) I_{D1}$$

$$I_{D1} = \frac{I_{in}}{1 + \frac{I_{S2}}{I_{S1}}}$$

$$\text{Similarly, } I_{D2} = \frac{I_{in}}{1 + \frac{I_{S1}}{I_{S2}}}$$

Robotics Application: Diode as a Protector

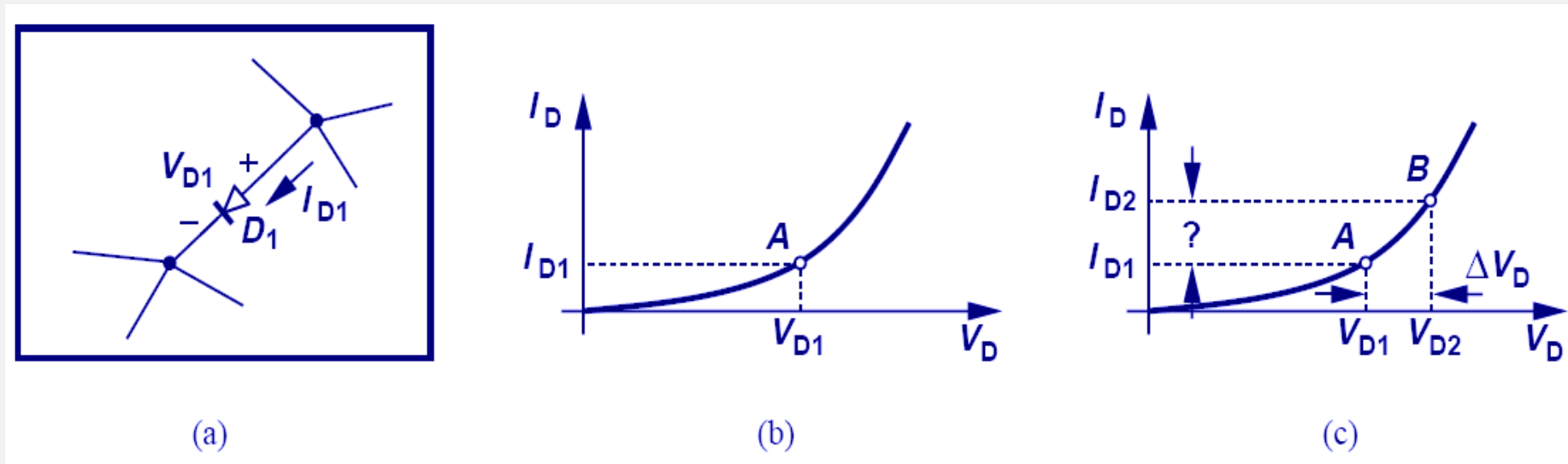


(a) Structure of a relay, (b) control of a motor by a relay, (c) problem of **voltage surge**, and (d) use of a diode to **protect** the microcontroller.

Small-Signal Analysis of Diode

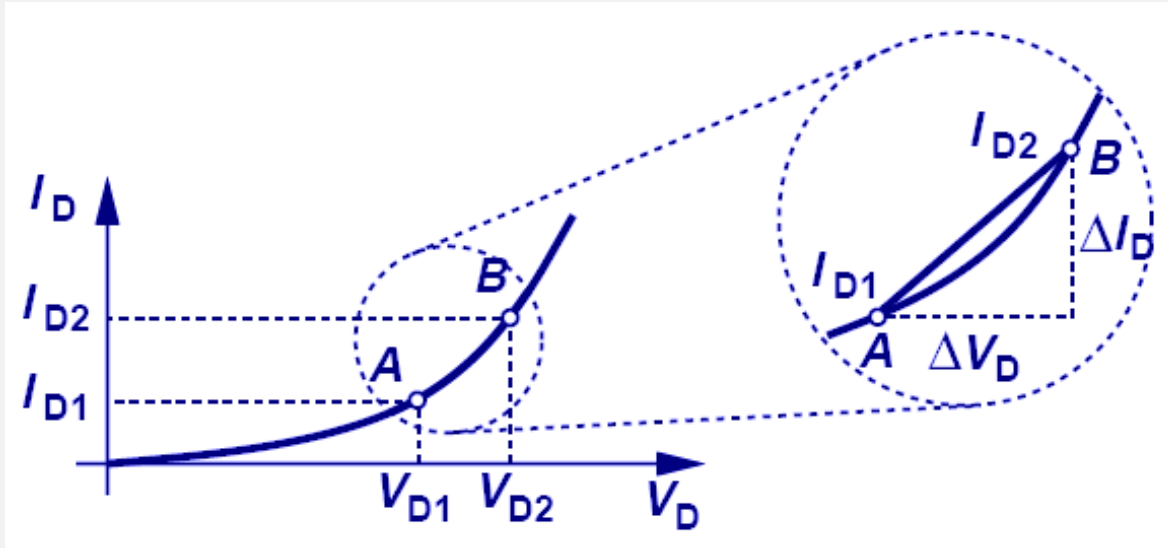
- Small-signal analysis is performed around a bias point by perturbing the voltage by a small amount and observing the resulting linear current perturbation.

$$\Delta V_D \rightarrow \Delta I_D?$$



(a) General circuit containing a diode, (b) operating point of D_1 (c) change in I_D as a result of change in V_D .

Small-Signal Analysis in Detail



Point A is called the “bias” point or the “operating” point.

$$I_{D2} = I_S \exp\left(\frac{V_{D2}}{V_T}\right)$$

$$= I_S \exp\left(\frac{V_{D1} + \Delta V_D}{V_T}\right)$$

$$= I_S \exp\left(\frac{V_{D1}}{V_T}\right) \times \exp\left(\frac{\Delta V_D}{V_T}\right)$$

$$\text{For } \frac{\Delta V_D}{V_T} \ll 1, \exp\left(\frac{\Delta V_D}{V_T}\right) \approx 1 + \frac{\Delta V_D}{V_T}.$$

$$= I_{D1} \left(1 + \frac{\Delta V_D}{V_T}\right)$$

$$= I_{D1} + I_{D1} \frac{\Delta V_D}{V_T}$$

$$\Delta I_D = I_{D2} - I_{D1} = I_{D1} \frac{\Delta V_D}{V_T}$$

$$\Delta I_D = \frac{\Delta V_D}{V_T} I_{D1}$$

Example 3-18

■ A diode is biased at a current of 1 mA. (a) Determine the current change if V_D changes by 1 mV. (b) Determine the voltage change if I_D changes by 10%.

Solution:

(a)

$$\begin{aligned}\Delta I_D &= \frac{\Delta V_D}{V_T} I_{D1} = \frac{1 \text{ mV}}{26 \text{ mV}} \times 1 \text{ mA} \\ &= 38.5 \text{ } \mu\text{A}\end{aligned}$$

(b)

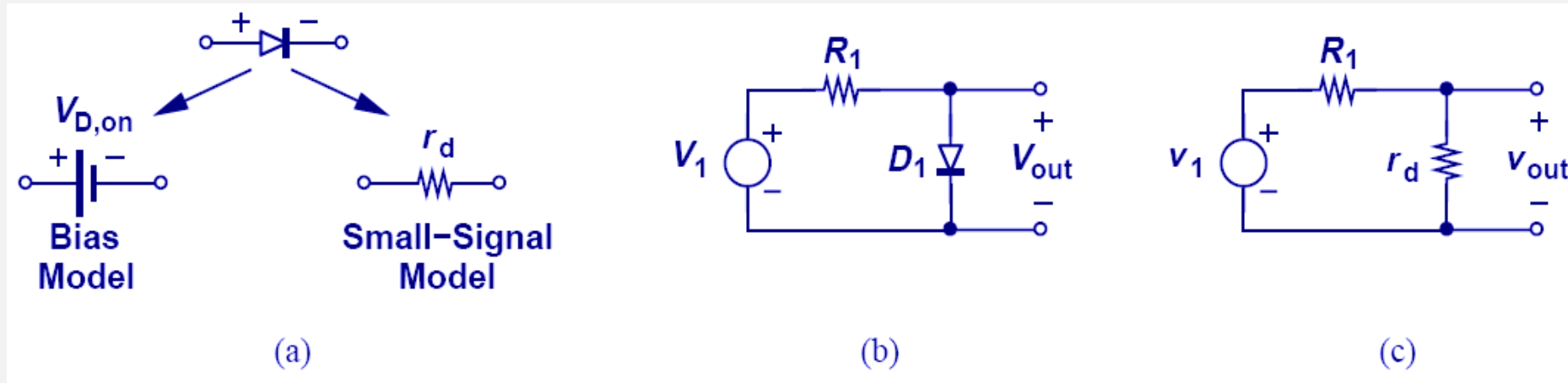
$$\begin{aligned}\Delta V_D &= \frac{V_T}{I_{D1}} \Delta I_D = \frac{26 \text{ mV}}{1 \text{ mA}} \times (0.1 \times 1 \text{ mA}) \\ &= 2.6 \text{ mV}\end{aligned}$$

Small-Signal Resistance

- In analogy with Ohm's law, we define the “**small-signal resistance**” of the diode as:

$$\Delta V_D = r_d \Delta I_D$$

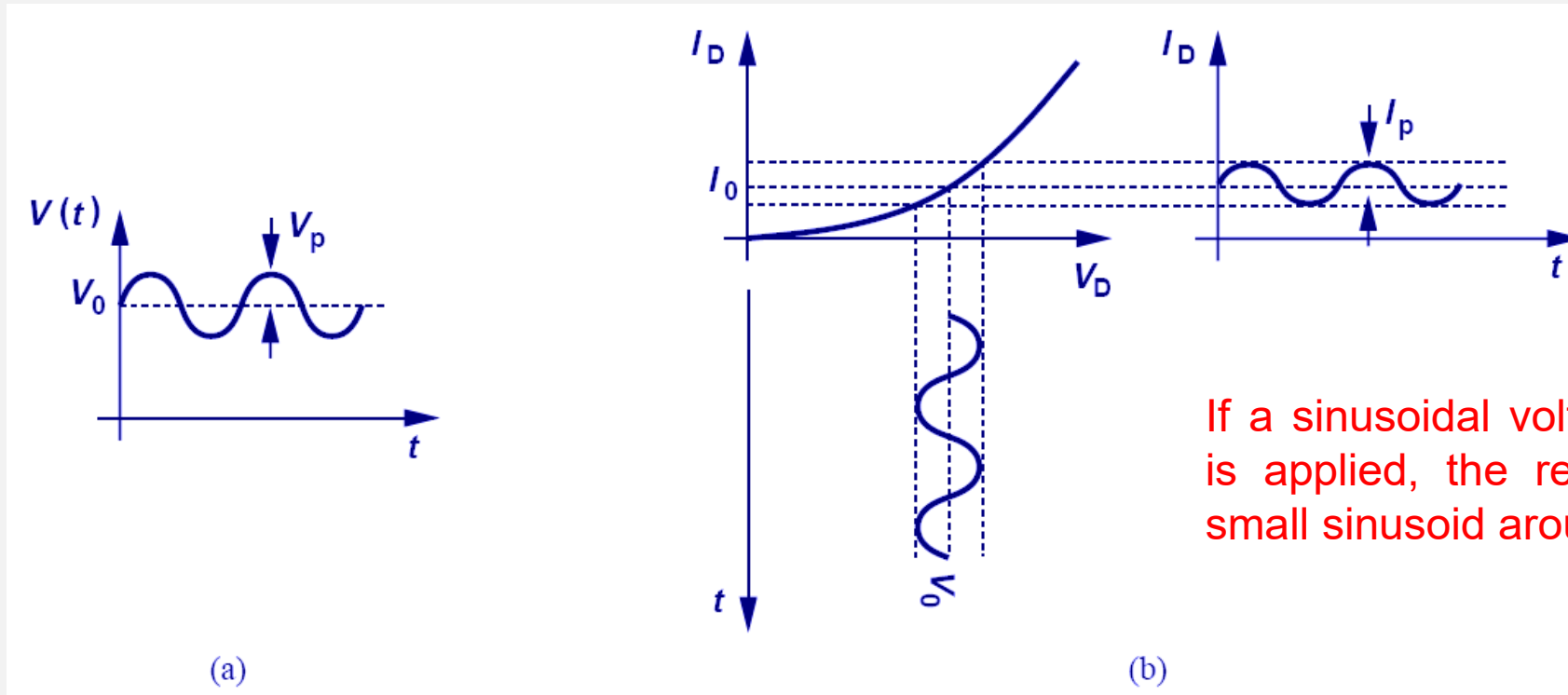
$$\text{where } r_d = \frac{V_T}{I_D}$$



(a) Summary of diode models for bias and signal calculations, (b) circuit example, (c) small-signal model.

Example 3-19: Small Signal Analysis

- A sinusoidal signal having a peak amplitude of V_p and a DC value of V_0 can be expressed as $V(t) = V_0 + V_p \cos \omega t$. If this signal is applied across a diode and $V_p \ll V_T$, determine the resulting **AC** current of the diode.



If a sinusoidal voltage with small amplitude is applied, the resulting current is also a small sinusoid around a DC value?

Solution:

Method #1:

$$\begin{aligned} I_D &= I_s \exp \frac{V_D}{V_T} = I_s \exp \frac{V_0 + V_p \cos \omega t}{V_T} \\ &= I_s \exp \frac{V_0}{V_T} \exp \frac{V_p \cos \omega t}{V_T} = I_0 \exp \frac{V_p \cos \omega t}{V_T} \\ &= I_0 \left(1 + \frac{V_p \cos \omega t}{V_T} \right) \quad (\because V_P \ll V_T) \\ &= I_0 + I_0 \frac{V_p \cos \omega t}{V_T} = I_0 + \frac{V_p \cos \omega t}{(V_T/I_0)} \\ \Delta I_D &= \frac{V_p \cos \omega t}{(V_T/I_0)} = \frac{V_p \cos \omega t}{r_d} = \frac{\Delta V_D}{r_d} \end{aligned}$$

✓ The AC current is also a sinusoidal signal.

Method #2:

Because $V_P \ll V_T$, we can apply small-signal analysis directly,

$$\Delta V_D = r_d \Delta I_D$$

$$\Delta I_D = \frac{\Delta V_D}{r_d} = \frac{V_p \cos \omega t}{r_d}$$

$$\text{where } r_d = \frac{V_T}{I_0} = \frac{V_T}{I_s \exp(V_0/V_T)}.$$

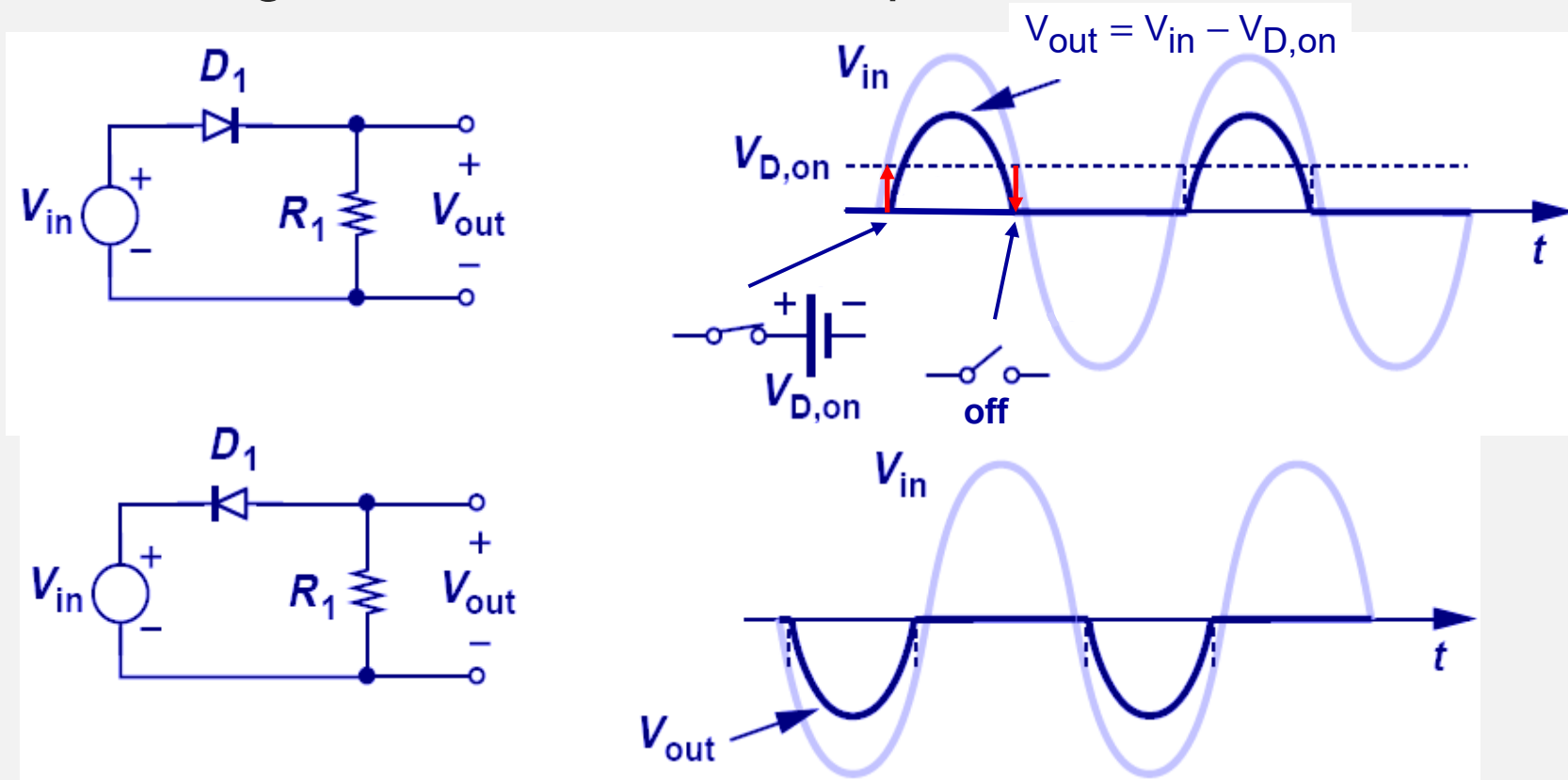
Applications of Diode

- The remainder of this chapter deals with circuit applications of diodes. A brief outline is shown in below.



Half-Wave Rectifier

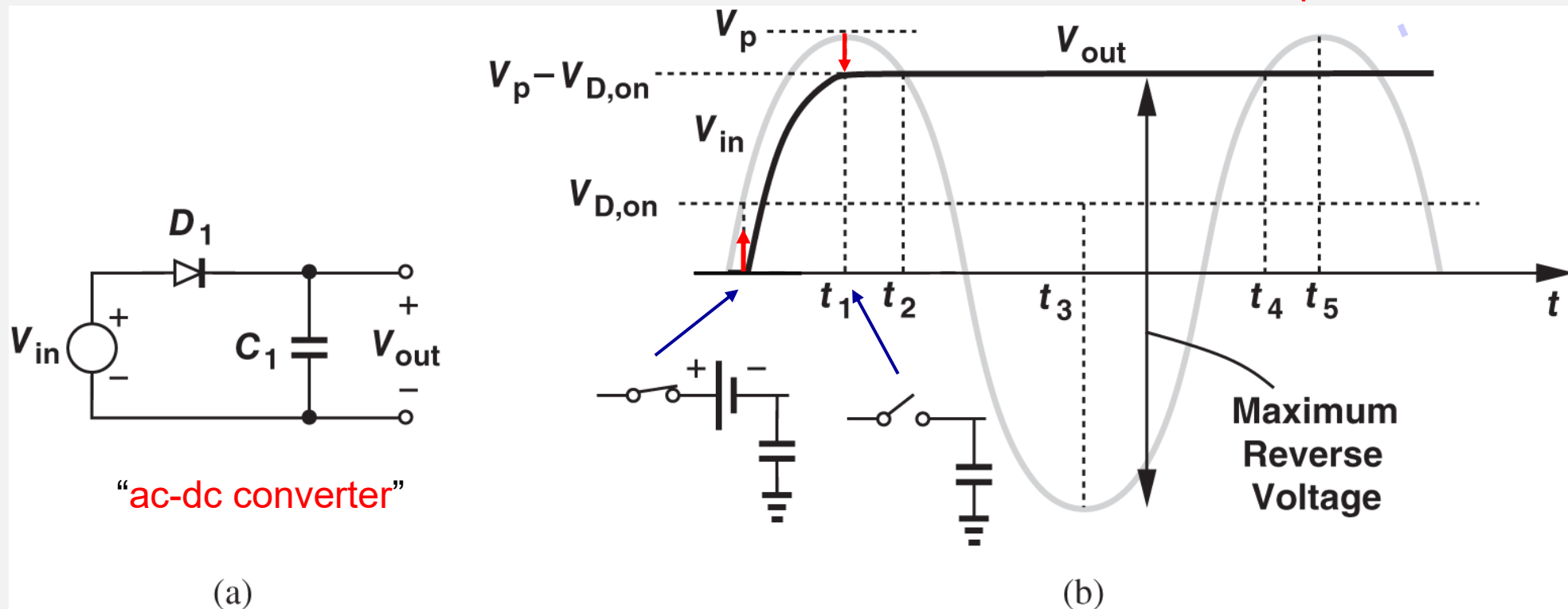
- A very common application of diodes is half-wave rectification, where either the positive or negative half of the input is blocked.
- But, how do we generate a *constant* output?



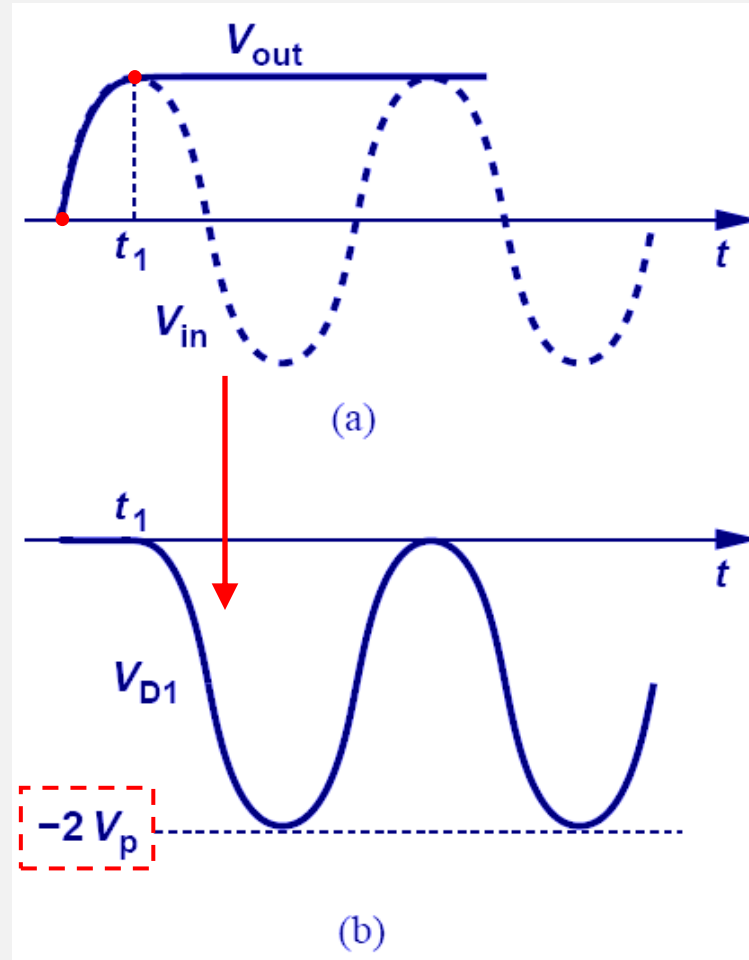
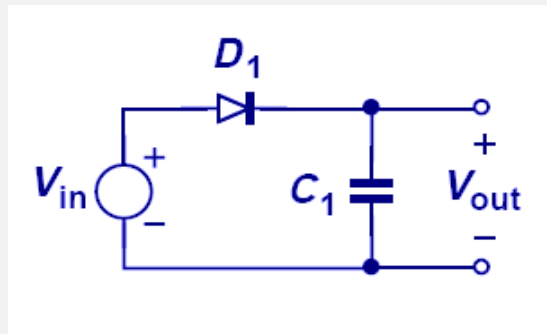
Diode-Capacitor Circuit: Constant Voltage Model

- If the resistor in half-wave rectifier is replaced by a capacitor, a fixed voltage output is obtained since the capacitor (assumed ideal) has no path to discharge.

This circuit is also called a “peak detector.”



Diode-Capacitor Circuit: Ideal Model



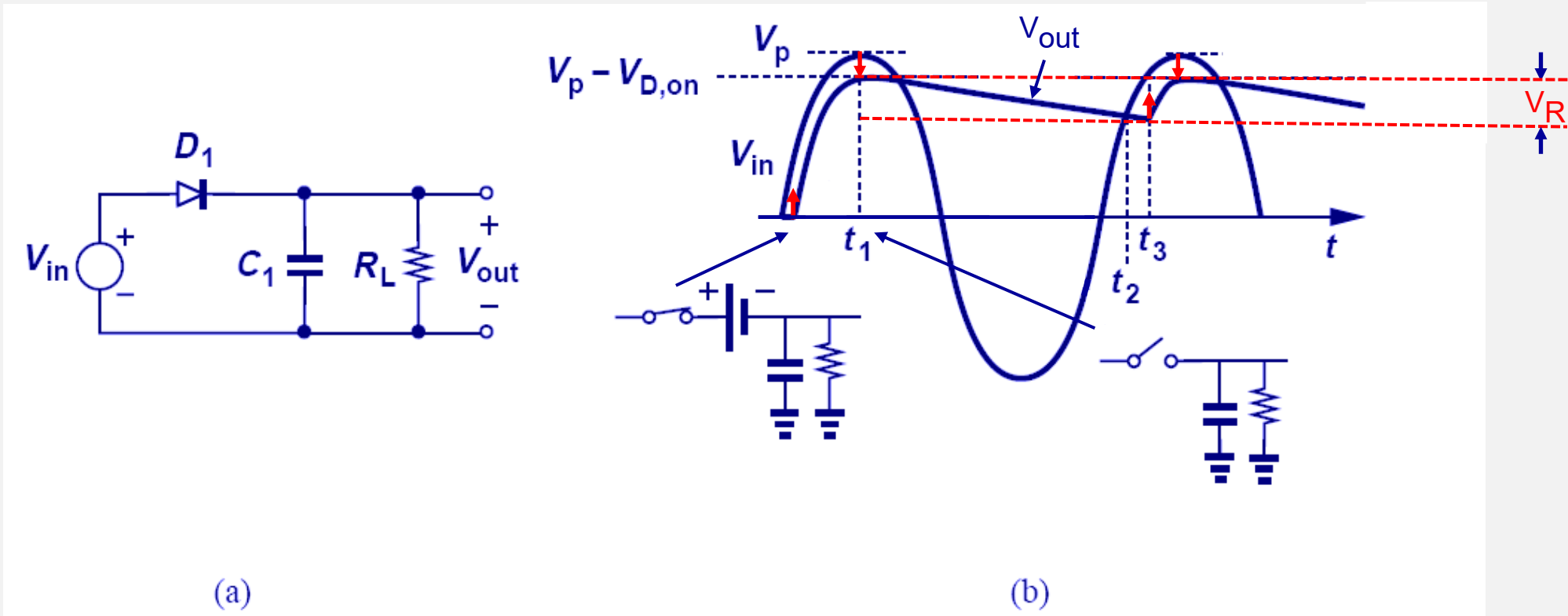
$$V_{D1} = V_{in} - V_{out}$$

Note that V_{D1} is just like V_{in} , only **shifted** down.

Diode-Capacitor with Load Resistor

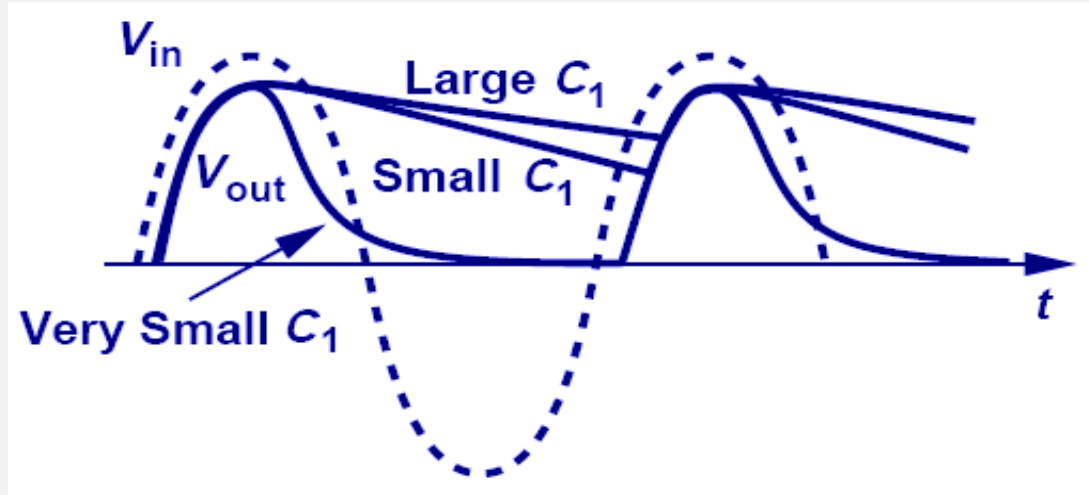
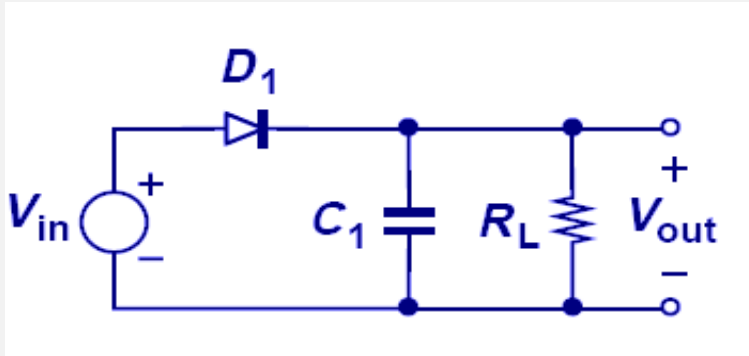
- A path is available for capacitor to discharge. Therefore, V_{out} will not be constant and a **ripple** exists.

The resulting variation in V_{out} is called the “**ripple**.”



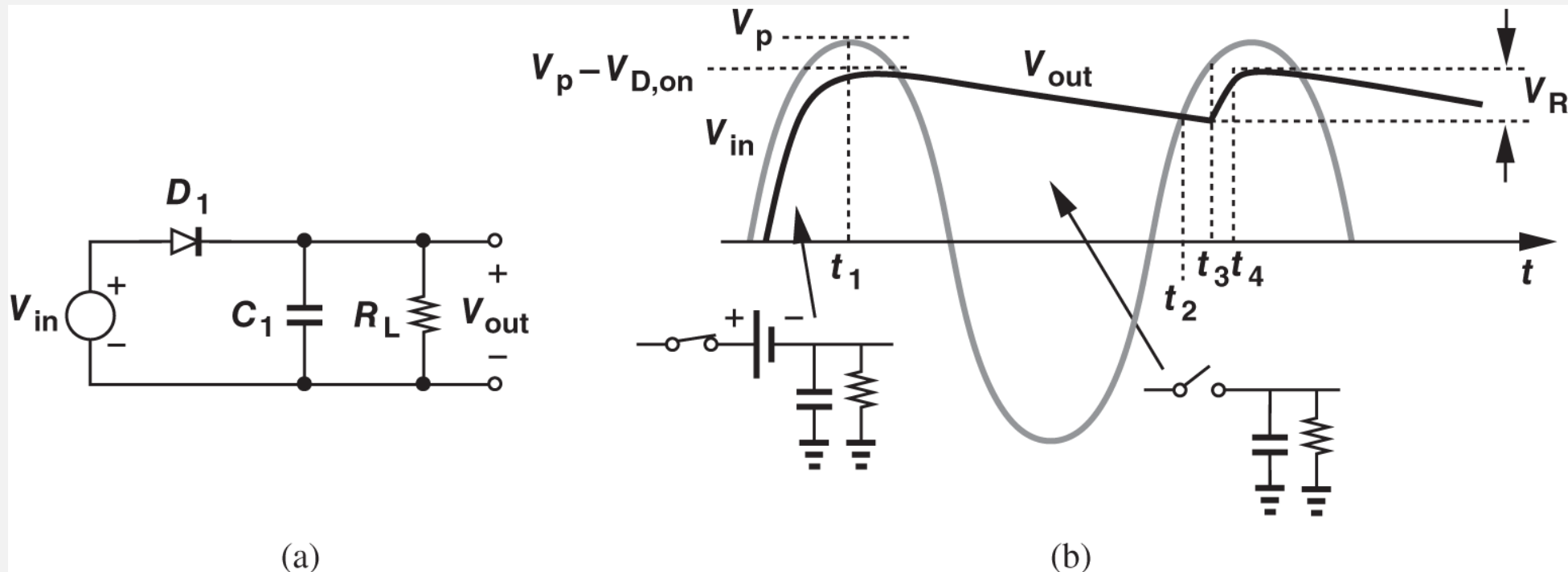
Behavior for Different Capacitor Values

- C_1 is called the “smoothing” or “filter” capacitor.
- For large C_1 , V_{out} has small ripple.



Peak to Peak Amplitude of Ripple

- In typical applications, the (peak-to-peak) amplitude of the ripple, V_R , must remain below 5% to 10% of the input peak voltage.
- If the maximum current drawn by the load is known, the value of C_1 is chosen large enough to yield an acceptable ripple.



Assume the frequency of V_{in} is f_{in} , i.e., the period $T_{in} = 1/f_{in}$. For simplicity, set the origin of time at t_1 , i.e., $t = t_1 = 0$. The discharge of C_1 through R_L can be expressed as

$$\begin{aligned} V_{out}(t) &= (V_p - V_{D,on}) \exp \frac{-t}{R_L C_1}, 0 \leq t \leq t_3 \\ &\approx (V_p - V_{D,on}) \left(1 - \frac{t}{R_L C_1} \right) \quad (\because \text{For small ripple } R_L C_1 \gg t_3) \\ &\approx (V_p - V_{D,on}) - \frac{V_p - V_{D,on}}{R_L} \frac{t}{C_1} \end{aligned}$$

$$\begin{aligned} V_R &= V_{out}(0) - V_{out}(t_3) \approx V_{out}(0) - V_{out}(T_{in}) \\ &\approx \frac{V_p - V_{D,on}}{R_L} \cdot \frac{T_{in}}{C_1} = I_L \cdot \frac{T_{in}}{C_1} = \frac{I_L}{C_1 f_{in}} \end{aligned}$$

Example 3-27

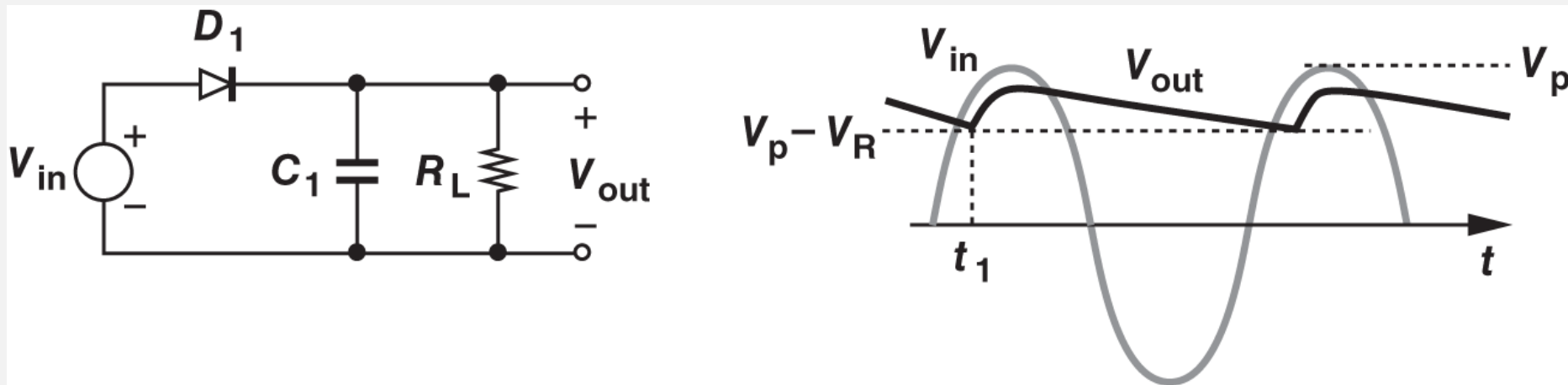
- A transformer converts the 110-V, 60-Hz line voltage to a peak-to-peak swing of 9 V. A half-wave rectifier follows the transformer to supply the power to the laptop computer of Example 3-25 ($R_L = 0.436$). Determine the minimum value of the filter capacitor that maintains the ripple below 0.1 V. Assume $V_{D,on} = 0.8$ V.

Solution:

$$\begin{aligned}\because V_R &\approx \frac{I_L}{C_1 f_{in}} \\ C_1 &\approx \frac{I_L}{V_R f_{in}} = \frac{V_p - V_{D,on}}{R_L} \cdot \frac{1}{V_R f_{in}} \\ &= \frac{\frac{9}{2} - 0.8}{0.436} \cdot \frac{1}{0.1 \times 60} = 1.417 \text{ F}\end{aligned}$$

Maximum Diode Current

- The diode has its maximum current at t_1 , since that's when the slope of V_{out} is the greatest.
- This current has to be carefully controlled so it does not **damage** the device.



Apply KCL at voltage output node,

$$I_{D1} = I_{C1} + I_L$$
$$I_{D1}(t) = C_1 \frac{dV_{out}}{dt} + I_L$$

Define the maximum diode current as I_p . Assume the ripple voltage is much less than the output voltage, i.e., $V_R \ll V_p$, I_L is approximated as a constant current.

$$I_p = \max_t \left\{ C_1 \frac{dV_{out}}{dt} + I_L \right\}$$
$$\approx C_1 \max_t \left\{ \frac{dV_{out}}{dt} \right\} + I_L \quad (1)$$

In other words, the maximum current of $I_{D1}(t)$ only depends on $I_{C1}(t)$ and $I_{D1}(t)$ reaches its maximum current when the slope of the output voltage $V_{out}(t)$ is maximum, i.e., $t = t_1$.

During the diode D_1 is on, $V_{out}(t) = V_{in}(t) - V_{D,on}$.

$$\begin{aligned} \frac{dV_{out}}{dt} &= \frac{dV_{in}}{dt} = \frac{d(V_p \sin \omega_{in} t)}{dt} \\ &= V_p \omega_{in} \cos \omega_{in} t \\ &= V_p \omega_{in} \sqrt{1 - \sin^2 \omega_{in} t} \end{aligned}$$

$$\max_t \left\{ \frac{dV_{out}}{dt} \right\} = V_p \omega_{in} \sqrt{1 - \sin \omega_{in} t_1^2} \quad (2)$$

$$\because V_{out}(t_1) = V_p - V_R$$

$$V_p \sin \omega_{in} t_1 = V_p - V_R$$

$$\sin \omega_{in} t_1 = 1 - \frac{V_R}{V_p} \quad (3)$$

From (1), (2) and (3),

$$\begin{aligned} I_p &= C_1 \max_t \left\{ \frac{dV_{out}}{dt} \right\} + I_L = C_1 V_p \omega_{in} \sqrt{1 - \left(1 - \frac{V_R}{V_p}\right)^2} + I_L \\ &= V_p C_1 \omega_{in} \sqrt{\frac{2V_R}{V_p} - \left(\frac{V_R}{V_p}\right)^2} + I_L \quad \left(\because V_R \ll V_p \Rightarrow \left(\frac{V_R}{V_p}\right)^2 \ll \frac{V_R}{V_p} \right) \\ &= V_p C_1 \omega_{in} \sqrt{\frac{2V_R}{V_p}} + I_L = V_p \frac{2\pi I_L}{V_R} \sqrt{\frac{2V_p}{V_R}} + I_L \quad \left(\because V_R \approx \frac{I_L}{C_1 f_{in}} \Rightarrow C_1 \omega_{in} \approx \frac{2\pi I_L}{V_R} \right) \\ &= I_L \left(1 + 2\pi \sqrt{\frac{2V_p}{V_R}} \right) \end{aligned}$$

Example 3-28

- Determine the peak diode current in Example 3-27 assuming $V_{D,on} \approx 0.8$ V and $C_1 = 1.417$ F. (Note: $V_{D,on} \approx 0$ is a typo in the textbook.)

Solution:

We have $V_p = 4.5$ V, $R_L = 0.436$ Ω , and $f_{in} = 60$ Hz.

$$I_L \approx \frac{V_p - V_{D,on}}{R_L} = \frac{4.5 - 0.8}{0.436} = 8.49 \text{ A}$$

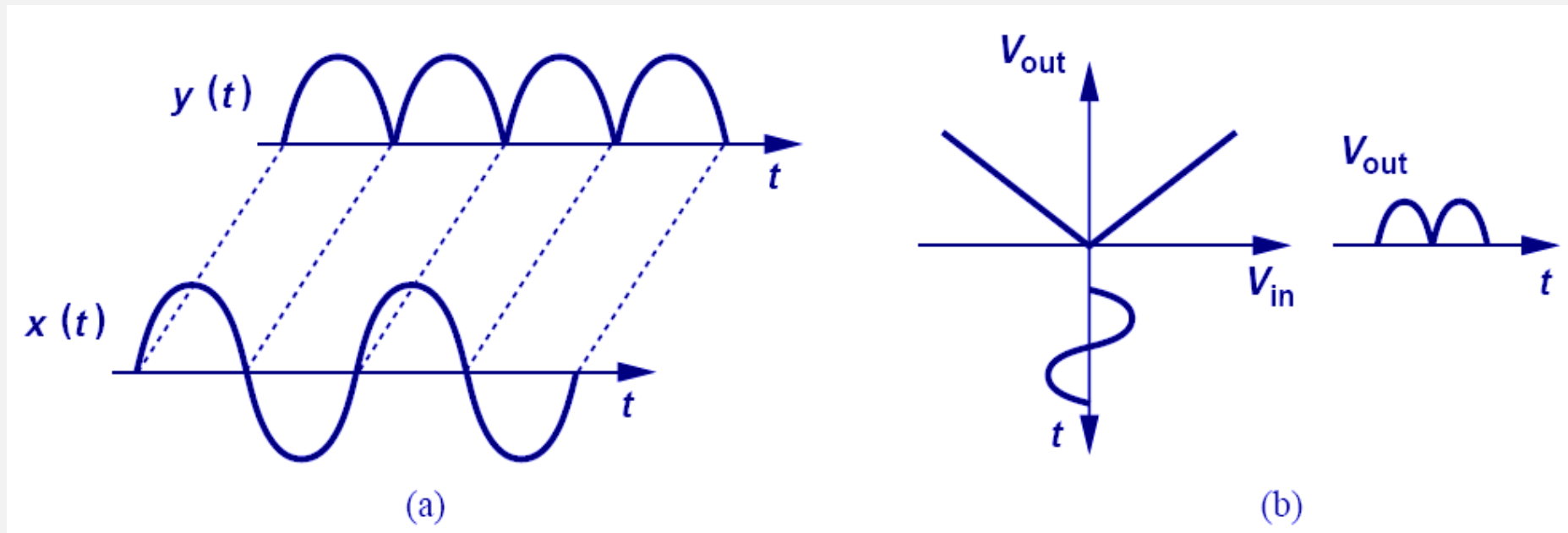
$$V_R \approx \frac{I_L}{C_1 f_{in}} = \frac{8.49}{1.417 \times 60} = 0.0998 \text{ V}$$

$$I_p = I_L \left(1 + 2\pi \sqrt{\frac{2V_p}{V_R}} \right) = 515 \text{ A}$$

Note: $V_R \approx 0.1$ V $\Rightarrow I_p = 517$ A

Full-Wave Rectifier (FWR)

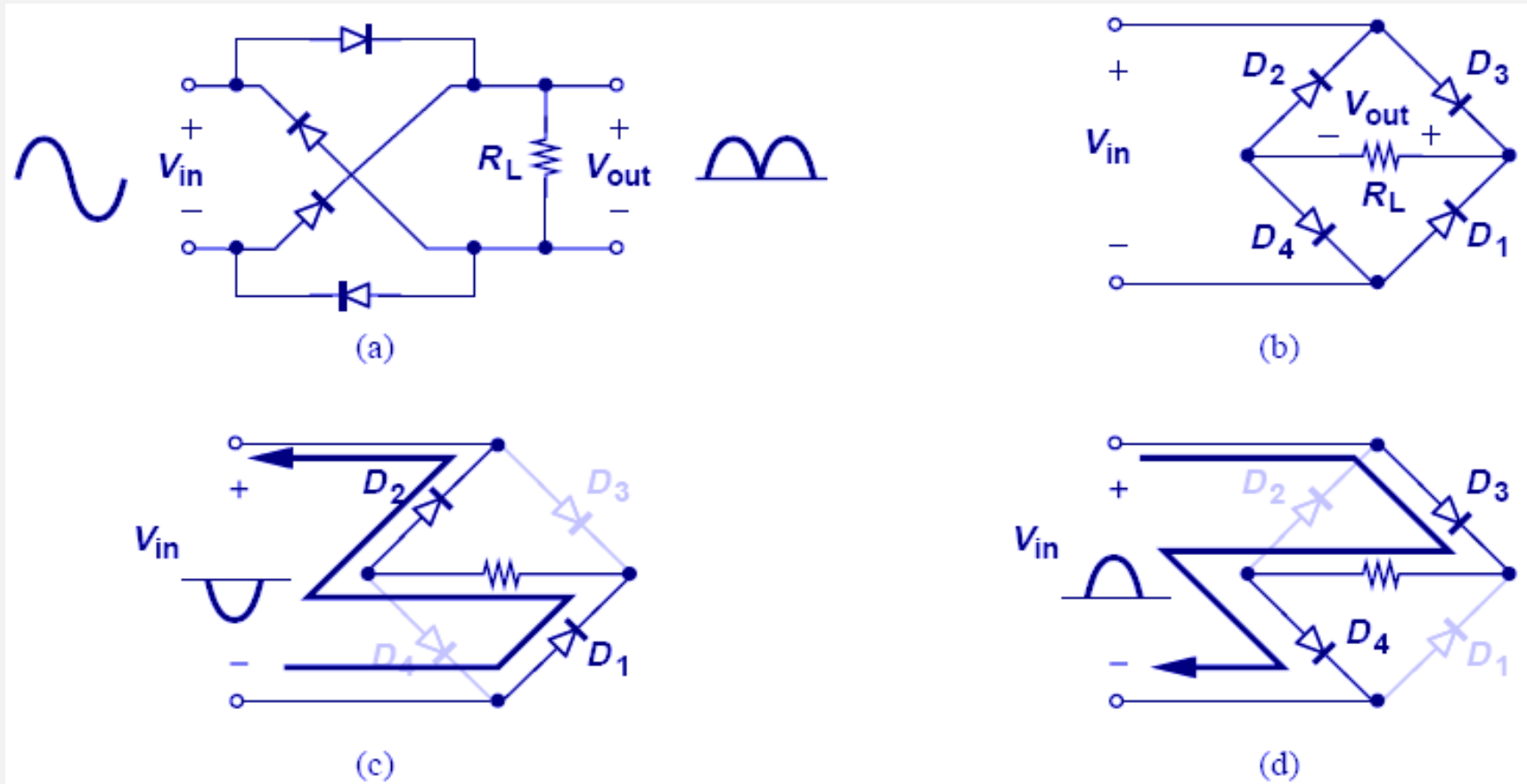
- A full-wave rectifier passes both the negative and positive half cycles of the input, while inverting the negative half of the input.
- As proved later, a full-wave rectifier reduces the ripple by a factor of two.



(a) Input and output waveforms and (b) input/output characteristic of a full-wave rectifier.

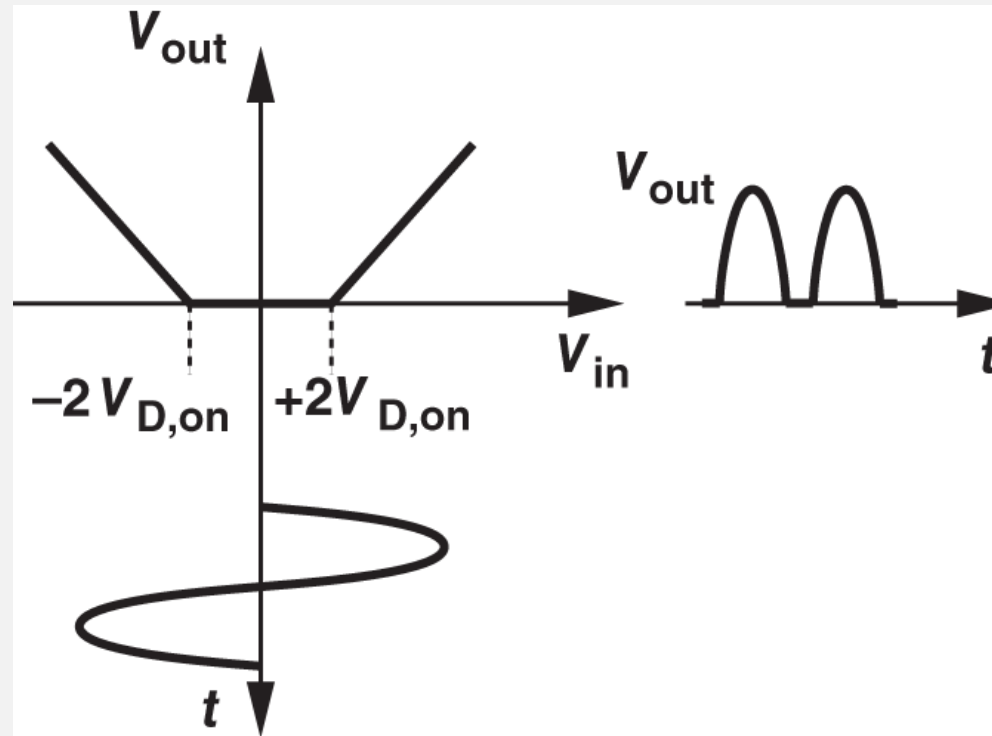
Full-Wave Rectifier: Bridge Rectifier

- The full-wave rectifier shown in (a) passes and invert the negative half cycles through D_1 and D_2 as in (c) and the positive half cycles through D_3 and D_4 as in (d).



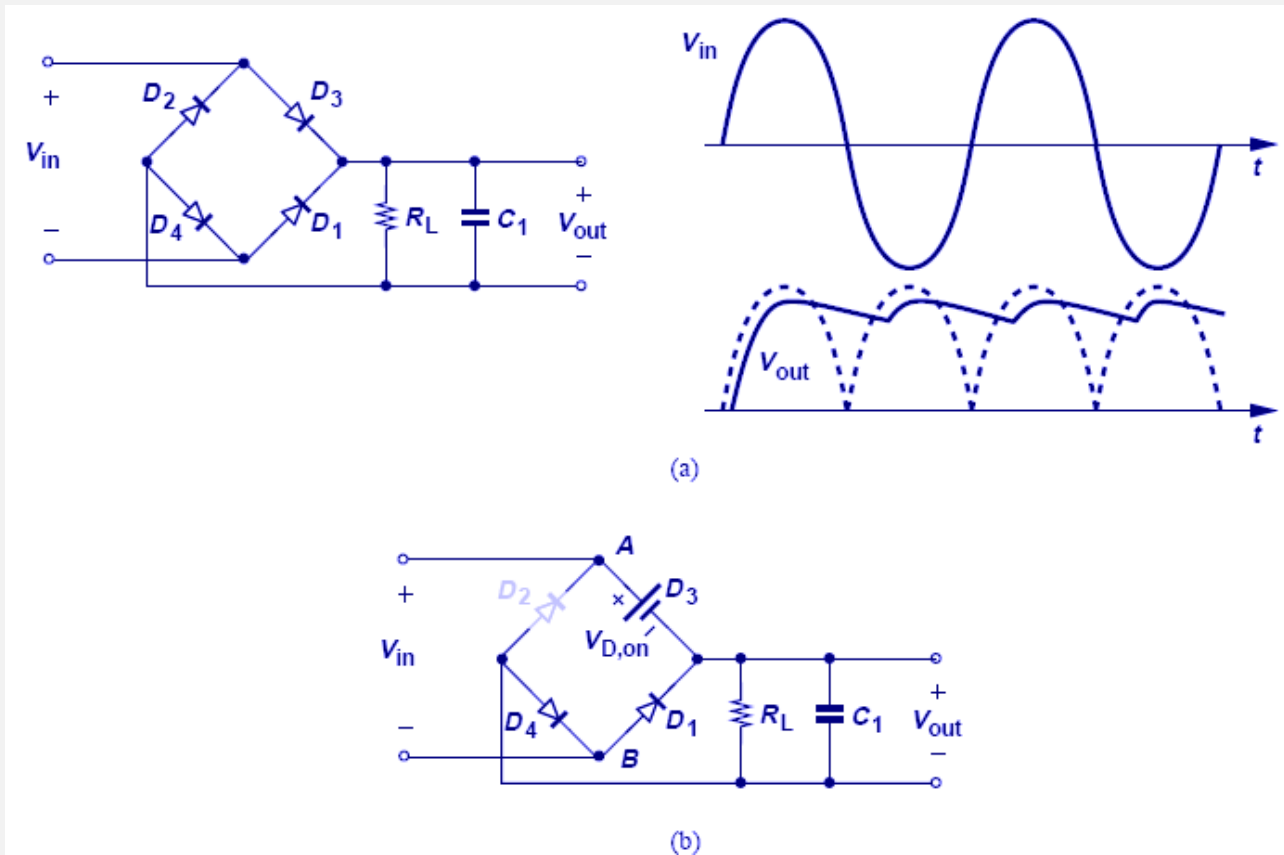
Input/Output Characteristics of a Full-Wave Rectifier with Constant-Voltage Model

- The dead-zone around V_{in} arises because V_{in} must exceed $2V_{D,ON}$ to turn on the bridge.



Complete Full-Wave Rectifier

- Since C_1 only gets $\frac{1}{2}$ of period to discharge, ripple voltage is decreased by a factor of 2. Also (b) shows that each diode is subjected to approximately one V_p reverse bias drop (versus $2V_p$ in half-wave rectifier).

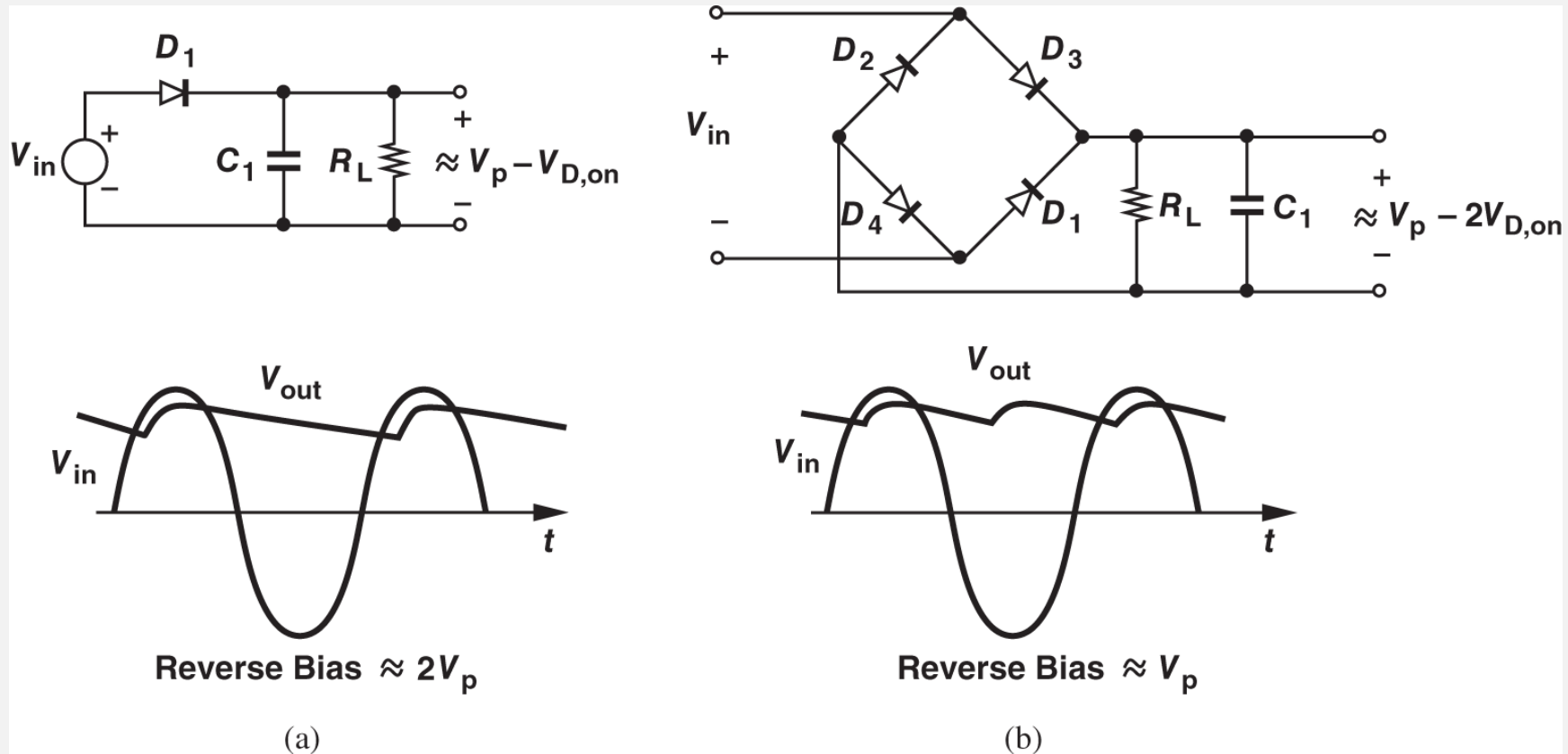


$$V_R \approx \frac{1}{2} \cdot \frac{V_p - 2V_{D,on}}{R_L C_1 f_{in}}$$

$$= \frac{1}{2} \cdot \frac{I_L}{C_1 f_{in}}$$

Advantages of Full-Wave Rectifiers

- By using two more diodes, full-wave rectifiers have half the ripple and require only half the diode breakdown voltage. Full-wave rectifier is more suited to adapter and charger applications.



Example 3-31

- Design a full-wave rectifier to deliver an average power of 2 W to a cellphone with a voltage of 3.6 V and a ripple of 0.2 V. Assume $V_{D,on} = 0.8$ V.

Solution:

We begin with the required input swing.

$$\begin{aligned} V_{in,p} &= V_{out} + 2V_{D,on} \\ &= 3.6 + 2 \times 0.8 = 5.2 \text{ V} \end{aligned}$$

Thus, the transformer preceding the rectifier must step the line voltage (110 V_{rms} or 220 V_{rms}) down to a peak value of 5.2 V.

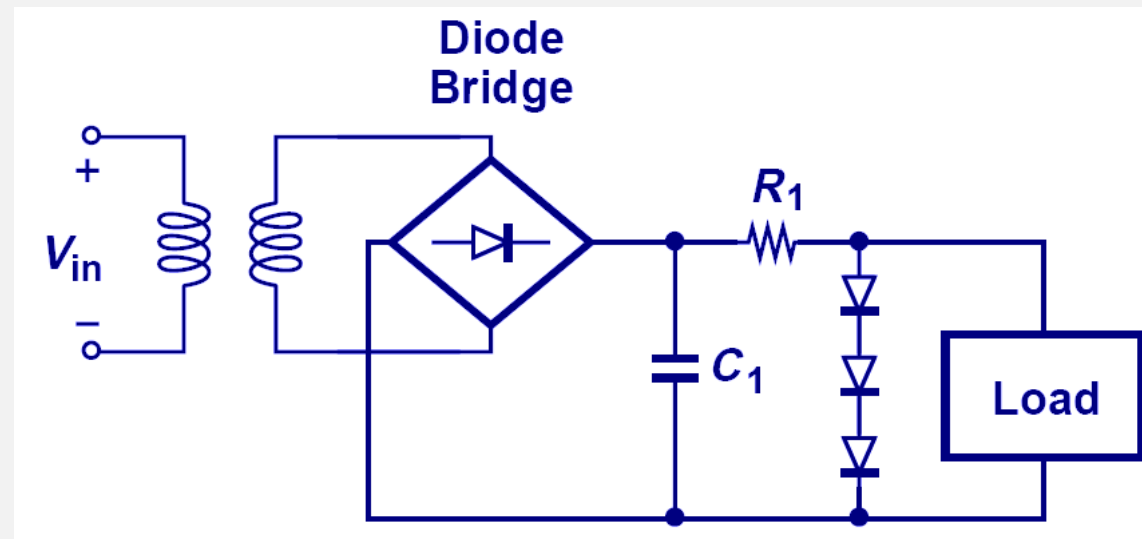
Next, we determine the minimum value of the smoothing capacitor that ensures $V_R \leq 0.2$ V.

$$\begin{aligned} \because V_R &= \frac{1}{2} \cdot \frac{I_L}{C_1 f_{in}} \\ C_1 &= \frac{1}{2} \cdot \frac{I_L}{V_R f_{in}} = \frac{1}{2} \cdot \frac{(P/V_{out})}{V_R f_{in}} = \frac{1}{2} \cdot \frac{(2 \text{ W}/3.6 \text{ V})}{(0.2 \text{ V}) \times (60 \text{ Hz})} = 23,100 \text{ } \mu\text{F} \end{aligned}$$

The diodes must withstand a reverse bias voltage of 5.2 V.

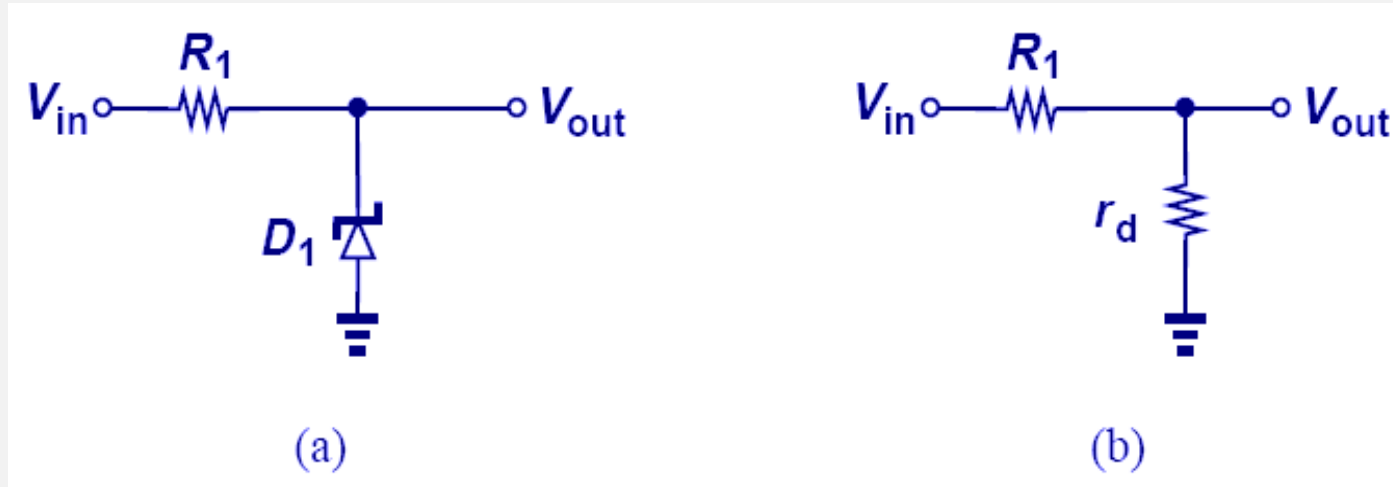
Voltage Regulator

- The ripple created by the rectifier can be unacceptable to sensitive load; therefore, a voltage regulator is required to obtain a very stable output.
- Three diodes operate as a primitive voltage regulator.



Voltage Regulation Using Zener Diode

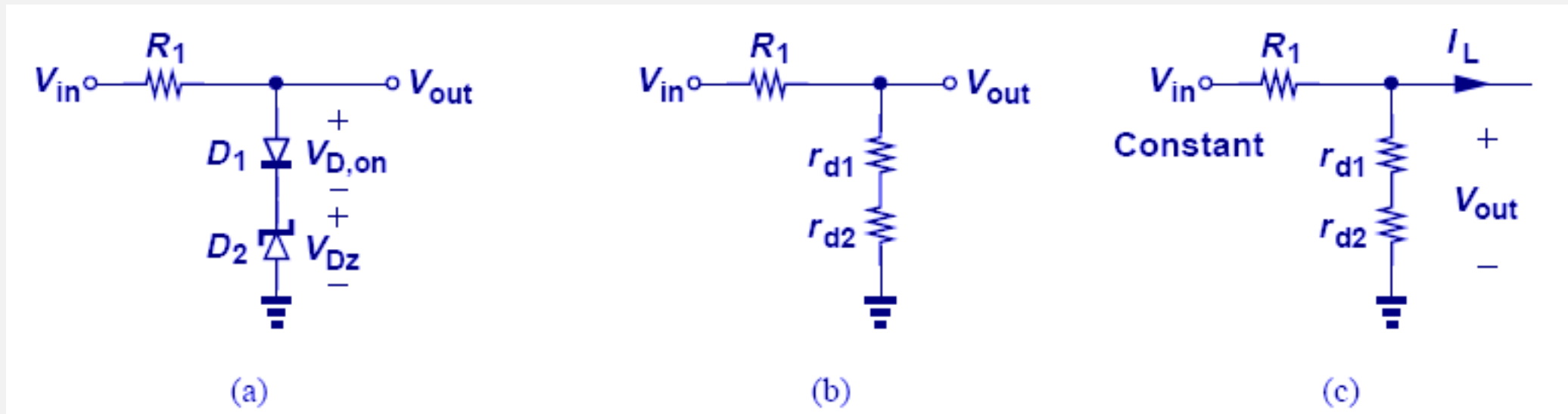
- Voltage regulation can be accomplished with Zener diode. Since r_d is small, large change in the input will not be reflected at the output.



$$V_{out} = \frac{r_d}{r_d + R_1} V_{in}$$

Line Regulation vs. Load Regulation

- **Line regulation:** the suppression of change in V_{out} due to change in input voltage V_{in} (b).
- **Load regulation:** the suppression of change in V_{out} due to change in load current I_L (c).



$$\frac{v_{out}}{v_{in}} = \frac{r_{d1} + r_{d2}}{r_{d1} + r_{d2} + R_1}$$

$$\left| \frac{v_{out}}{I_L} \right| = (r_{d1} + r_{d2}) || R_1$$

Example 3-33

- The circuit in previous page, V_{in} has a nominal value of 5 V, $R_1 = 100\ \Omega$, and D_2 has a reverse breakdown of 2.7 V and a small-signal resistance of $5\ \Omega$. Assuming $V_{D,on} \approx 0.8\text{ V}$ for D_1 determine the line and load regulation of the circuit.

Solution:

We first determine the bias current of and hence its small-signal resistance:

$$I_{D1} = \frac{V_{in} - V_{D1,on} - V_{DZ}}{R_1} = \frac{5 - 0.8 - 2.7}{100} = 15\text{ mA}$$
$$r_{D1} = \frac{V_T}{I_{D1}} = \frac{26\text{ mV}}{15\text{ mA}} = 1.73\ \Omega$$

The line regulation is

$$\frac{v_{out}}{v_{in}} = \frac{r_{D1} + r_{D2}}{r_{D1} + r_{D2} + R_1} = \frac{1.73 + 5}{1.73 + 5 + 100} = 0.0631$$

The load regulation is

$$\begin{aligned}\left|\frac{v_{out}}{I_L}\right| &= (r_{D1} + r_{D2}) \parallel R_1 \\ &= \frac{(r_{D1} + r_{D2})R_1}{(r_{D1} + r_{D2}) + R_1} \\ &= \frac{(1.73 + 5) \times 100}{(1.73 + 5) + 100} = 6.31\end{aligned}$$

Evolution of AC-DC Converter

